

 Miguel Ángel García Cortés

Nivel 1 – Fundamentos: imágenes y contenedores

1 Ejercicio 1.1 – Primer contacto con Alpine Linux Descarga y ejecuta un contenedor de Alpine Linux.

```
C:\Users\migue>docker pull alpine
Using default tag: latest
latest: Pulling from library/alpine
014e56e61396: Pull complete
Digest: sha256:51183f2cfa6320055da30872f211093f9ff1d3cf06f39a0bdb212314c5dc7375
Status: Downloaded newer image for alpine:latest
docker.io/library/alpine:latest
```

Comprueba su tamaño, entra en modo interactivo y muestra la versión del sistema.

```
C:\Users\migue>docker exec -it miguel_alpine /bin/sh
/ # cat /etc/alpine-release
3.23.0
/ #
```

```
C:\Users\migue>docker images alpine
REPOSITORY      TAG          IMAGE ID      CREATED        SIZE
alpine          latest       51183f2cfa63  8 days ago    13MB
```

2 Ejercicio 1.2 – Comprobando BusyBox Busca en Docker Hub la imagen oficial de BusyBox, descárgala, ejecútala y verifica qué comandos básicos tiene disponibles.

```
C:\Users\migue>docker search busybox --filter "is-official=true"
NAME            DESCRIPTION          STARS         OFFICIAL
busybox         Busybox base image.  3465         [OK]
```

```
C:\Users\migue>docker pull busybox
Using default tag: latest
latest: Pulling from library/busybox
Digest: sha256:d80cd694d3e9467884fcb94b8cale20437d8a501096cdf367a5a1918a34fc2fd
Status: Downloaded newer image for busybox:latest
docker.io/library/busybox:latest
```

```
C:\Users\migue>docker run -it --rm busybox
/ # busybox --help
BusyBox v1.37.0 (2024-09-26 21:31:42 UTC) multi-call binary.
BusyBox is copyrighted by many authors between 1998-2015.
Licensed under GPLv2. See source distribution for detailed
copyright notices.

Usage: busybox [function [arguments]...]
or: busybox --list[-full]
or: busybox --show SCRIPT
or: busybox --install [-s] [DIR]
or: function [arguments]...

    BusyBox is a multi-call binary that combines many common Unix
    utilities into a single executable. Most people will create a
    link to busybox for each function they wish to use and BusyBox
    will act like whatever it was invoked as.

Currently defined functions:
[, [[, acpid, add-shell, addgroup, adduser, adjtimex, ar, arch, arp, arping, ascii, ash, awk, base32, base64,
basename, bc, beep, blkdiscard, blkid, blockdev, bootchartd, brctl, bunzip2, bzip2, cal, cat, chat,
chattr, chgrp, chmod, chown, chpasswd, chpst, chroot, chrt, chvt, cksum, clear, cmp, comm, conspy, cp, cpio,
crc32, crond, crontab, cryptpw, ctttyhack, cut, date, dc, dd, deallocvt, delgroup, deluser, depmod, devmem, df,
dhcprelay, diff, dirname, dmesg, dnsd, dnsdomainname, dos2unix, dpkg, dpkg-deb, du, dumpkmap, dumpleases, echo,
ed, egrep, eject, env, envdir, envuidgid, ether-wake, expand, expr, factor, fakeidentd, fallocation, false,
fatattr, fbset, fbsplash, fdflush, fdformat, fdisk, fgconsole, fgrep, find, findfs, flock, fold, free,
freeramdisk, fsck, fsck.minix, fsfreeze, fstrim, fsync, ftpd, ftpget, ftpput, fuser, getfatattr, getopt, getty,
grep, groups, gunzip, gzip, halt, hd, hdparm, head, hexdump, hexedit, hostid, hostname, httpd, hush, hwclock,
i2cdetect, i2cdump, i2cget, i2cset, i2ctransfer, id, ifconfig, ifdown, ifenslave, ifplugd, ifup, inetd, init,
inotifyd, insmod, install, ionice, iostat, ip, ipaddr, ipcalc, ipcrm, ipcs, iplink, ipneigh, iproute, iprule,
iptunnel, kbd_mode, kill, killall, klogd, last, less, link, linux32, linux64, linuxrc, ln, loadfont,
loadkmap, logger, login, logname, logread, losetup, lpd, lpq, lpr, ls, lsattr, lsmod, lsof, lspci, lsscsi,
```

3 Ejercicio 1.3 – Contenedor CentOS Descarga la imagen oficial de CentOS y crea un contenedor que permanezca activo.

```
C:\Users\migue>docker run -d --name miguel_centos centos:7 sleep infinity
1ae9bc825ae5ce6bf7590df116dc75e9cb09ec156df8df6da0424df25c88853b
```

Verifica su versión y deténlo

```
C:\Users\migue>docker exec miguel_centos cat /etc/centos-release
CentOS Linux release 7.9.2009 (Core)
```

```
C:\Users\migue>docker stop miguel_centos
miguel_centos
```

4 Ejercicio 1.4 – Explorando contenedores Lista todos los contenedores (en ejecución o detenidos), inspecciona uno para ver su dirección IP interna, y elimina aquellos que ya no necesitas.

```
C:\Users\migue>docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED         STATUS              PORTS          NAMES
44b1df75fd7c   busybox    "sleep infinity"        15 minutes ago  Up 15 minutes                      miguel_busybox
1ae9bc825ae5   centos:7   "sleep infinity"        17 minutes ago  Exited (137) 13 minutes ago         miguel_centos
a90c47aldab1   alpine     "/bin/sh"               53 minutes ago  Up 41 minutes                      miguel_alpine
```

```
SecondaryIPV6Addresses": null,
"EndpointID": "70d4b31763af7fa6c37082508a5ec72dba524d00c8f4392a91345da503110d52",
"Gateway": "172.17.0.1",
"GlobalIPv6Address": "",
"GlobalIPv6PrefixLen": 0,
"IPAddress": "172.17.0.4",
"IPPrefixLen": 16,
"IPv6Gateway": "",
"MacAddress": "32:6d:52:6a:46:d2",
"Networks": {
```

```
C:\Users\migue>docker stop miguel_alpine
miguel_alpine

C:\Users\migue>docker stop miguel_centos
miguel_centos

C:\Users\migue>docker stop miguel_busybox
miguel_busybox

C:\Users\migue>docker rm miguel_alpine
miguel_alpine

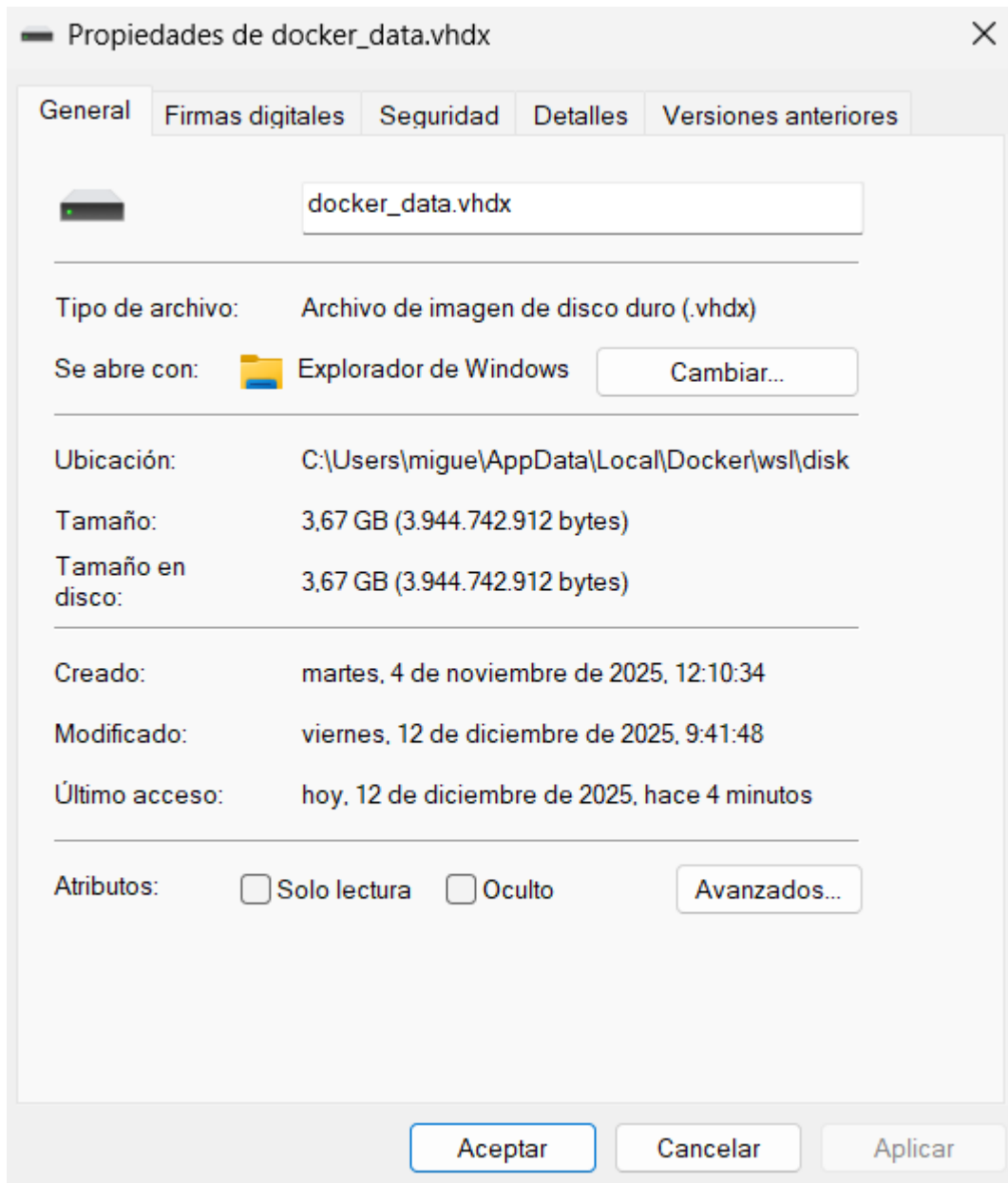
C:\Users\migue>docker rm miguel_centos
miguel_centos

C:\Users\migue>docker rm miguel_busybox
miguel_busybox

C:\Users\migue>
```

[5] Ejercicio 1.5 – Información del sistema Docker Muestra información detallada de tu instalación Docker (versiones, almacenamiento, redes, etc.) y localiza el directorio donde se guardan las imágenes.

```
C:\Users\migue>docker info
Client:
Version:      28.5.1
Context:      desktop-linux
Debug Mode:   false
Plugins:
ai: Docker AI Agent - Ask Gordon (Docker Inc.)
   Version:   v1.9.11
   Path:      C:\Program Files\Docker\cli-plugins\docker-ai.exe
buildx: Docker Buildx (Docker Inc.)
   Version:   v0.29.1-desktop.1
   Path:      C:\Program Files\Docker\cli-plugins\docker-buildx.exe
compose: Docker Compose (Docker Inc.)
   Version:   v2.40.3-desktop.1
   Path:      C:\Users\migue\.docker\cli-plugins\docker-compose.exe
debug: Get a shell into any image or container (Docker Inc.)
   Version:   0.0.45
   Path:      C:\Program Files\Docker\cli-plugins\docker-debug.exe
desktop: Docker Desktop commands (Docker Inc.)
   Version:   v0.2.0
   Path:      C:\Program Files\Docker\cli-plugins\docker-desktop.exe
extension: Manages Docker extensions (Docker Inc.)
   Version:   v0.2.31
   Path:      C:\Program Files\Docker\cli-plugins\docker-extension.exe
init: Creates Docker-related starter files for your project (Docker Inc.)
   Version:   v1.4.0
   Path:      C:\Program Files\Docker\cli-plugins\docker-init.exe
```



○ Nivel 2 – Redes docker

6 Ejercicio 2.1 – Creando una red personalizada Crea una red llamada `red_lab` y lanza en ella tres contenedores de diferentes distribuciones pequeñas (alpine, busybox, centos).

```
C:\Users\migue>docker network create red_lab
a48f51e2343513a53ca64cfb7308eb7327b15c2b090f814ea472a0633b0cf6ea
```

```
C:\Users\migue>docker run -d --name miguel_alpine --network red_lab alpine sleep infinity
ccleb5f3cec9385de7370726a70791115b2861362f0075a953a012891e731d54
```

```
C:\Users\migue>docker run -d --name miguel_busybox --network red_lab busybox sleep infinity
688305bc458e1445c2e582117598c52ddb0a820b5ccaa668f9cea7ad6c9a5591
```

```
C:\Users\migue>docker run -d --name miguel_centos --network red_lab centos:7 sleep infinity
c1cb956d876febd71ed1740da13832a2f0e5db5759661c39459f3483d837c7b0
```

Comprueba que pueden hacerse ping entre sí por nombre.

```
C:\Users\migue>docker exec miguel_alpine ping -c 2 miguel_busybox
PING miguel_busybox (172.18.0.3): 56 data bytes
64 bytes from 172.18.0.3: seq=0 ttl=64 time=0.190 ms
64 bytes from 172.18.0.3: seq=1 ttl=64 time=0.066 ms

--- miguel_busybox ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.066/0.128/0.190 ms

C:\Users\migue>docker exec miguel_alpine ping -c 2 miguel_centos
PING miguel_centos (172.18.0.4): 56 data bytes
64 bytes from 172.18.0.4: seq=0 ttl=64 time=0.176 ms
64 bytes from 172.18.0.4: seq=1 ttl=64 time=0.060 ms

--- miguel_centos ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.060/0.118/0.176 ms

C:\Users\migue>docker exec miguel_centos ping -c 2 miguel_busybox
PING miguel_busybox (172.18.0.3) 56(84) bytes of data.
64 bytes from miguel_busybox.red_lab (172.18.0.3): icmp_seq=1 ttl=64 time=0.109 ms
64 bytes from miguel_busybox.red_lab (172.18.0.3): icmp_seq=2 ttl=64 time=0.051 ms

--- miguel_busybox ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1030ms
rtt min/avg/max/mdev = 0.051/0.080/0.109/0.029 ms

C:\Users\migue>
```

7 Ejercicio 2.2 – Un contenedor en dos redes Crea una segunda red llamada red_extra y conecta uno de los contenedores anteriores a ambas redes.

Comprueba su configuración de red interna.

```
C:\Users\migue>docker network create red_extra
b98538ad9eb97a6f1bd3b45dff8f0602dcacf00066f277fd98eb02034d2d52234
```

```
C:\Users\migue>docker network connect red_extra miguel_alpine
```

```
C:\Users\migue>docker network inspect red_extra
[
  {
    "Name": "red_extra",
    "Id": "b98538ad9eb97a6f1bd3b45dff8f0602dcacf00066f277fd98eb02034d2d52
234",
    "Created": "2025-12-15T18:02:40.663827551Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.19.0.0/16",
          "Gateway": "172.19.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {
      "cc1eb5f3cec9385de7370726a70791115b2861362f0075a953a012891e731d5
4": {
        "Name": "miguel_alpine",
        "EndpointID": "b90d2177bd38a327f0aaf9af3e0f4e2ac23e07d21e07c
ee2c80cf2f0f6868d63",
        "MacAddress": "ca:65:dc:f1:12:95",
        "IPv4Address": "172.19.0.2/16",
        "IPv6Address": ""
      }
    }
  }
]
```

8 Ejercicio 2.3 – Servidor HTTP temporal Dentro de un contenedor python:3-alpine, inicia un servidor web con python3 -m http.server 8080 y accede a él desde otro contenedor de la misma red usando curl.

```
C:\Users\migue>docker run -d --name servidor_web --network red_lab python:3-alpine python3 -m http.server 8080
```

```
/ # curl -v http://servidor_web:8080
* Host servidor_web:8080 was resolved.
* IPv6: (none)
* IPv4: 172.18.0.5
* Trying 172.18.0.5:8080...
* Established connection to servidor_web (172.18.0.5 port 8080) from 172.18.0.6 port 35312
* using HTTP/1.x
> GET / HTTP/1.1
> Host: servidor_web:8080
> User-Agent: curl/8.17.0
> Accept: */*
```

🌀 Nivel 3 – Volúmenes y persistencia

📁 Ejercicio 3.1 – Volumen compartido entre contenedores.

Crea un volumen llamado compartido y monta dicho volumen en dos contenedores distintos (alpine y busybox).

```
C:\Users\migue>docker volume create compartido
compartido

C:\Users\migue>docker run -it --rm -v compartido:/datos alpine /bin/sh
/ # |
```

Guarda un archivo en uno y verifica que aparece en el otro.

```
C:\Users\migue>docker run -it --rm -v compartido:/datos alpine /bin/sh
/ # cd /datos
/datos # echo "Hola desde el contenedor Alpine" > mensaje.txt
/datos # ls -l
total 4
-rw-r--r--    1 root    root           32 Dec 15 18:17 mensaje.txt
/datos # exit

C:\Users\migue>docker run -it --rm -v compartido:/datos busybox
/ # cat /datos/mensaje.txt
Hola desde el contenedor Alpine
/ # |
```

📁 Ejercicio 3.2 – Volumen persistente con Alpine Lanza un contenedor alpine montando un volumen llamado vol_asir en la carpeta /datos.

Dentro del contenedor crea un archivo llamado mensaje.txt con el texto Hola ASIR.

Elimina el contenedor y lanza otro nuevo utilizando el mismo volumen.

Comprueba que el archivo mensaje.txt sigue existiendo y conserva su contenido original.

```
C:\Users\migue>docker run -it --rm -v vol_asir:/datos alpine /bin/sh
/ # echo "Hola ASIR" > /datos/mensaje.txt
/ # ls -l /datos
total 4
-rw-r--r--    1 root    root          10 Dec 15 18:41 mensaje.txt
/ # exit

C:\Users\migue>docker run -it --rm -v vol_asir:/datos alpine /bin/sh
/ # cat /datos/mensaje.txt
Hola ASIR
/ # exit

C:\Users\migue>
```


● Nivel 4 – Mantenimiento y limpieza

11 Ejercicio 4.1 – Limpieza y diagnóstico del entorno Lista imágenes, contenedores, volúmenes y redes.

Elimina los elementos que no estén en uso.

Comprueba cuánto espacio ocupa Docker en tu sistema antes y después.

```
C:\Users\migue>docker system df
```

TYPE	TOTAL	ACTIVE	SIZE	RECLAIMABLE
Images	13	5	1.637GB	1.264GB (77%)
Containers	4	4	6.59MB	0B (0%)
Local Volumes	2	0	42B	42B (100%)
Build Cache	0	0	0B	0B

```
C:\Users\migue>docker system prune -a --volumes
WARNING! This will remove:
 - all stopped containers
 - all networks not used by at least one container
 - all anonymous volumes not used by at least one container
 - all images without at least one container associated to them
 - all build cache

Are you sure you want to continue? [y/N] y
Deleted Networks:
redC
redA
redB

Deleted Images:
untagged: ubuntu:25.04
deleted: sha256:27771fb7b40a58237c98e8d3e6b9ecdd9289cec69a857fccfb85ff36294d
ac20
deleted: sha256:34e8533bf27ac50f60bec267f6ce18c9aeb9556574e1ec1a8ce89926d32e
a8f3
untagged: rockylinux/rockylinux:latest
deleted: sha256:fc370d748f4cd1e6ac3d1b6460fb82201897fa15a16f43e947940df5aca1
a56e
deleted: sha256:2f0bf3347b762fb21264670b046758782673694883cdf031af3aba982f65
6830
untagged: debian:latest
deleted: sha256:01a723bf5bfb21b9dda0c9a33e0538106e4d02cce8f557e118dd61259553
d598
deleted: sha256:75a32e5b9e4aecc3454bab811b66fb98c8a44679915cf93f7d47370f6a7f
b643
```

```
C:\Users\migue>docker system df
```

TYPE	TOTAL	ACTIVE	SIZE	RECLAIMABLE
Images	4	4	387.4MB	11.96MB (3%)
Containers	4	4	6.59MB	0B (0%)
Local Volumes	2	0	42B	42B (100%)
Build Cache	0	0	0B	0B